

Solutions – Final Exam

ECON 320 – Introduction to Mathematical Economics

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Instructions:

- You are allowed **120 minutes** to complete the exam.
 - Show all work. Unsupported answers receive little credit.
 - Unless stated otherwise, all variables are real and differentiable as needed.
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Multiple Choice Questions (20 points)

Each question is worth 2 points. Correct options are indicated and briefly justified.

1. $Q = \sqrt{P + 10} - I$.

$$\frac{\partial Q}{\partial I} = -1.$$

Answer: (b) -1 .

2. $z = x^2y^3$.

$$\begin{aligned} z_x &= 2xy^3, & z_y &= 3x^2y^2, \\ dz &= z_x dx + z_y dy = 2xy^3 dx + 3x^2y^2 dy. \end{aligned}$$

Answer: (a).

3. $f(x) = 4x - x^3$.

$$f'(x) = 4 - 3x^2 = 0 \Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \pm\sqrt{\frac{4}{3}}.$$

Second derivative:

$$f''(x) = -6x.$$

At $x = \sqrt{4/3} > 0$, $f''(x) < 0 \Rightarrow$ local maximum.

Answer: (a) local maximum.

4. $Y = AK^{0.5}L^{0.5}$.

In Cobb–Douglas, elasticity of Y w.r.t. K equals the exponent on K , here 0.5.

Answer: (a) 0.5.

5. $f(x, y) = 3x^2 + 4xy + y^2$.

$$f_{xx} = 6, \quad f_{yy} = 2, \quad f_{xy} = f_{yx} = 4,$$

$$\det(H) = 6 \cdot 2 - 4^2 = 12 - 16 = -4 < 0,$$

so the stationary point (if any) is a saddle.

Answer: (a) saddle point.

6. $\max U(x, y) = xy$ s.t. $x + y = 10$.

The Lagrange multiplier λ measures the marginal increase in maximum utility if the right-hand side of the constraint (10) is increased: shadow value of relaxing the constraint.

Answer: (c) marginal utility of relaxing the constraint.

7. $\max_x f(x, a) = ax - x^2$.

Envelope theorem:

$$\frac{dv}{da} = \left. \frac{\partial f}{\partial a} \right|_{x=x^*(a)} = x^*.$$

Answer: (a) x .

8. Demand $P(Q) = 20 - Q$, price $P = 10$.

At $P = 10$, $10 = 20 - Q \Rightarrow Q = 10$. Consumer surplus is area of triangle:

$$\text{CS} = \frac{1}{2}(\text{choke price} - P) \cdot Q = \frac{1}{2}(20 - 10) \cdot 10 = \frac{1}{2} \cdot 10 \cdot 10 = 50.$$

Answer: (b) 50.

9. $\frac{dy}{dt} = 4y$, $y(0) = 3$.

Solve:

$$y(t) = Ce^{4t}, \quad y(0) = 3 \Rightarrow C = 3 \Rightarrow y(t) = 3e^{4t}.$$

Answer: (b) $y = 3e^{4t}$.

10. $Y(t) = K^{0.3}L^{0.7}e^{0.02t}$ with K, L constant.

$$\frac{dY}{dt} = K^{0.3}L^{0.7} \cdot 0.02e^{0.02t} = 0.02Y.$$

Answer: (a) $\frac{dY}{dt} = 0.02Y$.

Problems (80 points total)

Problem 1 (16 points)

A firm produces

$$Q = x^{1/2}y^{1/3}.$$

(a) Compute dQ . (5 pts)

Partial derivatives:

$$Q_x = \frac{\partial Q}{\partial x} = \frac{1}{2}x^{-1/2}y^{1/3}, \quad Q_y = \frac{\partial Q}{\partial y} = \frac{1}{3}x^{1/2}y^{-2/3}.$$

Total differential:

$$dQ = Q_x dx + Q_y dy = \frac{1}{2}x^{-1/2}y^{1/3} dx + \frac{1}{3}x^{1/2}y^{-2/3} dy.$$

(b) If the firm controls only x , compute dQ/dx . (3 pts)

Holding y fixed:

$$\frac{dQ}{dx} = Q_x = \frac{1}{2}x^{-1/2}y^{1/3}.$$

(c) Profit $\pi(x) = 40Q - x - y - 5$, find optimal x (with y fixed). (5 pts)

Substitute Q :

$$\pi(x) = 40x^{1/2}y^{1/3} - x - y - 5.$$

Differentiate w.r.t. x (treating y as constant):

$$\pi'(x) = 40 \cdot \frac{1}{2}x^{-1/2}y^{1/3} - 1 = 20x^{-1/2}y^{1/3} - 1.$$

FOC:

$$20x^{-1/2}y^{1/3} - 1 = 0 \quad \Rightarrow \quad 20x^{-1/2}y^{1/3} = 1.$$

Solve for x . Rearrange:

$$x^{-1/2} = \frac{1}{20y^{1/3}} \quad \Rightarrow \quad x^{1/2} = 20y^{1/3}.$$

Squaring:

$$x^* = (20y^{1/3})^2 = 400y^{2/3}.$$

(d) Maximum or minimum? (3 pts)

Second derivative:

$$\pi''(x) = \frac{d}{dx} (20x^{-1/2}y^{1/3} - 1) = 20y^{1/3} \cdot \left(-\frac{1}{2}x^{-3/2}\right) = -10y^{1/3}x^{-3/2}.$$

For $x > 0$ and $y > 0$, $x^{-3/2} > 0$, $y^{1/3} > 0$, hence

$$\pi''(x^*) < 0,$$

so π is locally concave and x^* is a local maximum (and global maximum on $x > 0$).

Problem 2 (16 points)

$$R(t) = 500e^{0.03t}(1 + t^{1/2}).$$

(a) Compute $\frac{dR}{dt}$. (5 pts)

Write $R(t) = 500u(t)v(t)$ with

$$u(t) = e^{0.03t}, \quad v(t) = 1 + t^{1/2}.$$

Then

$$u'(t) = 0.03e^{0.03t}, \quad v'(t) = \frac{1}{2}t^{-1/2}.$$

Product rule:

$$\frac{dR}{dt} = 500(u'v + uv') = 500 \left[0.03e^{0.03t}(1 + t^{1/2}) + e^{0.03t}\frac{1}{2}t^{-1/2} \right].$$

Factor:

$$\frac{dR}{dt} = 500e^{0.03t} \left[0.03(1 + t^{1/2}) + \frac{1}{2}t^{-1/2} \right].$$

(b) Growth rate $g(t) = \frac{1}{R} \frac{dR}{dt}$. (4 pts)

$$R(t) = 500e^{0.03t}(1 + t^{1/2}),$$

so

$$g(t) = \frac{\frac{dR}{dt}}{R} = \frac{500e^{0.03t} \left[0.03(1 + t^{1/2}) + \frac{1}{2}t^{-1/2} \right]}{500e^{0.03t}(1 + t^{1/2})}.$$

Cancel the common factor:

$$g(t) = \frac{0.03(1 + t^{1/2}) + \frac{1}{2}t^{-1/2}}{1 + t^{1/2}} = 0.03 + \frac{1}{2} \frac{t^{-1/2}}{1 + t^{1/2}}.$$

(c) Elasticity of R w.r.t. time. (4 pts)

Elasticity:

$$E_{R,t} = \frac{dR}{dt} \cdot \frac{t}{R} = g(t) t.$$

Hence

$$E_{R,t}(t) = t \left[0.03 + \frac{1}{2} \frac{t^{-1/2}}{1 + t^{1/2}} \right] = 0.03t + \frac{t^{1/2}}{2(1 + t^{1/2})}.$$

(d) Accelerating or decelerating growth? (3 pts)

Observe:

$$g(t) = 0.03 + \frac{1}{2} \frac{t^{-1/2}}{1 + t^{1/2}}.$$

The second term is positive for $t > 0$ but *decreases* as t increases, and

$$\lim_{t \rightarrow \infty} g(t) = 0.03.$$

So the growth rate declines toward a constant 3%. Revenue growth is *decelerating*, approaching a steady growth rate.

Problem 3 (16 points)

$$f(x, y) = 20x - 4x^2 + 10y - 3y^2 - xy.$$

(a) First-order partial derivatives. (4 pts)

$$f_x = 20 - 8x - y, \quad f_y = 10 - 6y - x.$$

(b) Solve the first-order conditions. (4 pts)

Set $f_x = f_y = 0$:

$$20 - 8x - y = 0 \quad \Rightarrow \quad y = 20 - 8x,$$

$$10 - 6y - x = 0 \quad \Rightarrow \quad 6y + x = 10 \Rightarrow y = \frac{10 - x}{6}.$$

Equate the two expressions for y :

$$20 - 8x = \frac{10 - x}{6}.$$

Multiply both sides by 6:

$$120 - 48x = 10 - x \quad \Rightarrow \quad 120 - 10 = 48x - x \quad \Rightarrow \quad 110 = 47x.$$

Thus

$$x^* = \frac{110}{47}.$$

Then

$$y^* = 20 - 8x^* = 20 - 8 \left(\frac{110}{47} \right) = \frac{20 \cdot 47 - 8 \cdot 110}{47} = \frac{940 - 880}{47} = \frac{60}{47}.$$

(c) Hessian and determinant at the critical point. (4 pts)

Second derivatives:

$$f_{xx} = -8, \quad f_{yy} = -6, \quad f_{xy} = f_{yx} = -1.$$

Hessian:

$$H = \begin{bmatrix} -8 & -1 \\ -1 & -6 \end{bmatrix}.$$

Determinant:

$$\det(H) = (-8)(-6) - (-1)^2 = 48 - 1 = 47 > 0.$$

(d) Classification. (4 pts)

We have $f_{xx} = -8 < 0$ and $\det(H) > 0$, which implies the Hessian is negative definite. Therefore, $(x^*, y^*) = \left(\frac{110}{47}, \frac{60}{47} \right)$ is a strict local maximum. Since f is a quadratic with a negative definite Hessian everywhere, this is also the unique global maximum.

Problem 4 (16 points)

$$U(x, y) = x^{1/2}y^{1/2}, \quad p_x x + p_y y = I.$$

(a) Lagrangian and FOCs. (5 pts)

Lagrangian:

$$\mathcal{L}(x, y, \lambda) = x^{1/2}y^{1/2} - \lambda(p_x x + p_y y - I).$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2}x^{-1/2}y^{1/2} - \lambda p_x = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2}x^{1/2}y^{-1/2} - \lambda p_y = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(p_x x + p_y y - I) = 0 \quad \Rightarrow \quad p_x x + p_y y = I.$$

(b) Solve for x^* and y^* . (5 pts)

Divide the first FOC by the second (eliminate λ):

$$\frac{\frac{1}{2}x^{-1/2}y^{1/2}}{\frac{1}{2}x^{1/2}y^{-1/2}} = \frac{\lambda p_x}{\lambda p_y} \Rightarrow \frac{y/x}{1} = \frac{p_x}{p_y}.$$

On the left:

$$\frac{x^{-1/2}y^{1/2}}{x^{1/2}y^{-1/2}} = x^{-1}y = \frac{y}{x}.$$

Hence:

$$\frac{y}{x} = \frac{p_x}{p_y} \Rightarrow y = \frac{p_x}{p_y}x.$$

Use the budget constraint:

$$p_x x + p_y y = I \Rightarrow p_x x + p_y \left(\frac{p_x}{p_y} x \right) = p_x x + p_x x = 2p_x x = I,$$

so

$$x^* = \frac{I}{2p_x}.$$

Then

$$y^* = \frac{p_x}{p_y} x^* = \frac{p_x}{p_y} \cdot \frac{I}{2p_x} = \frac{I}{2p_y}.$$

(c) Use the envelope theorem to compute $\partial V / \partial I$. (3 pts)

The indirect utility function is

$$V(p_x, p_y, I) = U(x^*, y^*) = \left(\frac{I}{2p_x} \right)^{1/2} \left(\frac{I}{2p_y} \right)^{1/2} = \frac{I}{2\sqrt{p_x p_y}}.$$

By the envelope theorem for this equality-constrained maximization,

$$\frac{\partial V}{\partial I} = \lambda^*,$$

where λ^* is the optimal Lagrange multiplier.

We can compute directly from V :

$$\frac{\partial V}{\partial I} = \frac{1}{2\sqrt{p_x p_y}}.$$

$$\text{Thus } \lambda^* = \frac{1}{2\sqrt{p_x p_y}}.$$

(d) Interpretation. (3 pts)

$\partial V / \partial I = \lambda^*$ is the *marginal utility of income*: the increase in maximum attainable utility from a one-unit increase in income. It is positive and decreasing in the geometric mean of prices $\sqrt{p_x p_y}$: higher prices reduce the marginal value of additional income.

Problem 5 (16 points)

Capital evolves as

$$\frac{dK}{dt} = I(t) - \delta K, \quad I(t) = 50e^{0.01t}, \quad \delta = 0.05.$$

(a) Solve the differential equation for $K(t)$. (6 pts)

Rewrite in standard linear form:

$$\frac{dK}{dt} + 0.05K = 50e^{0.01t}.$$

Integrating factor:

$$\mu(t) = e^{\int 0.05 dt} = e^{0.05t}.$$

Multiply both sides by $\mu(t)$:

$$e^{0.05t} \frac{dK}{dt} + 0.05e^{0.05t} K = 50e^{0.05t} e^{0.01t} = 50e^{0.06t}.$$

Left-hand side is derivative of $Ke^{0.05t}$:

$$\frac{d}{dt}(Ke^{0.05t}) = 50e^{0.06t}.$$

Integrate:

$$K(t)e^{0.05t} = \int 50e^{0.06t} dt = \frac{50}{0.06}e^{0.06t} + C = \frac{2500}{3}e^{0.06t} + C.$$

Thus

$$K(t) = \frac{2500}{3}e^{0.06t}e^{-0.05t} + Ce^{-0.05t} = \frac{2500}{3}e^{0.01t} + Ce^{-0.05t}.$$

(b) Particular solution with $K(0) = 100$. (4 pts)

At $t = 0$,

$$K(0) = \frac{2500}{3} \cdot 1 + C \cdot 1 = \frac{2500}{3} + C = 100.$$

Hence

$$C = 100 - \frac{2500}{3} = \frac{300 - 2500}{3} = -\frac{2200}{3}.$$

So

$$K(t) = \frac{2500}{3}e^{0.01t} - \frac{2200}{3}e^{-0.05t}.$$

(c) Compute $\int_0^{20} I(t) dt$. (3 pts)

$$\int_0^{20} I(t) dt = \int_0^{20} 50e^{0.01t} dt = 50 \int_0^{20} e^{0.01t} dt = 50 \left[\frac{1}{0.01} e^{0.01t} \right]_0^{20} = 50 \cdot 100(e^{0.2} - 1).$$

So

$$\int_0^{20} I(t) dt = 5000(e^{0.2} - 1).$$

(d) Long-run behavior of $K(t)$. (3 pts)

From

$$K(t) = \frac{2500}{3}e^{0.01t} - \frac{2200}{3}e^{-0.05t},$$

as $t \rightarrow \infty$,

$$e^{0.01t} \rightarrow \infty, \quad e^{-0.05t} \rightarrow 0.$$

Thus

$$K(t) \sim \frac{2500}{3}e^{0.01t} \rightarrow \infty.$$

Economically: because investment grows over time at a positive rate (1%) and dominates depreciation in the long run, the capital stock keeps increasing without bound, growing asymptotically at net rate 1%.